

Week 3 Worksheet 1 Solutions

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Exercise 1. The hamiltonian for the harmonic oscillator is

$$H = \frac{1}{2m}[p^2 + (m\omega x)^2].$$

a) Show that

$$H = \hbar\omega(a^\dagger a + 1/2),$$

where

$$\begin{aligned}a^\dagger &= \alpha x + i\beta p \\ a &= \alpha x - i\beta p.\end{aligned}$$

That is, determine α and β .

b) Show that $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$.

c) Show that there is a solution ψ_0 to the TISE such that $a^\dagger a \psi_0 = 0$.

Hints: First, show that $[N, a] = -a$. If $N = a^\dagger a$ and $N\psi = \lambda\psi$ for some scalar λ and some normalizable ψ , show that $Na\psi = (\lambda - 1)a\psi$. Then, show that $|a\psi|^2 = \psi^* N\psi$. What does this tell you about repeated applications of a to ψ ?

d) What is the energy of ψ_0 ?

e) Show that $N = a^\dagger a$ is the **number operator**: If $\psi_n = (a^\dagger)^n \psi_0$ is the n^{th} stationary state, then

$$N\psi_n = n\psi_n.$$

f) Show that if $N\psi_0 = 0$, then a complete set of solutions to the equation $N\psi = \lambda\psi$ is given by $(a^\dagger)^n \psi_0$, so that $(a^\dagger)^n \psi_0$ are also a complete set of solutions to the TISE.

a) We compute

$$a^\dagger a = \alpha^2 x^2 + \beta^2 p^2 + \hbar\alpha\beta.$$

From this we can read off that

$$|\alpha| = \sqrt{\frac{m\omega}{2\hbar}}$$

$$|\beta| = \frac{1}{\sqrt{2m\hbar\omega}}.$$

To determine the signs, we can solve the equation

$$\hbar\alpha\beta + \frac{1}{2} = 0.$$

This has solution

$$\alpha\beta = -\frac{1}{2\hbar}.$$

If we want to keep α positive, then we set

$$\beta = -\frac{1}{\sqrt{2m\hbar\omega}}.$$

b) We compute

$$[a, a^\dagger] = -2\hbar\alpha\beta = 1.$$

c) First we need to check that $[N, a] = -a$. We compute

$$\begin{aligned} [N, a] &= a^\dagger a^2 - a a^\dagger a \\ &= [a^\dagger, a]a \\ &= -a. \end{aligned}$$

Then

$$Na\psi = (-a + aN)\psi = (\lambda - 1)a\psi.$$

To show that

$$|a\psi|^2 = \psi^* N\psi,$$

it's sufficient to check that $a^* = a^\dagger$, which is obvious from the definitions. This means that repeated applications of a lower the eigenvalue of ψ under N . Now, if $|a^n\psi|^2 = (a^{n-1}\psi)^* Na^{n-1}\psi = (\lambda - n - 1)|a^{n-1}\psi|^2$, that means that there will be some large n such that $\lambda - n - 1 \leq 0$. Since $|a^n\psi|^2 \geq 0$, we find that $\lambda - n - 1 = 0$ for some large n . Thus, there is a state with $\lambda' = \lambda - n - 1 = 0$.

d) It's $\hbar\omega/2$, since $N\psi_0 = 0$.

e) We compute

$$N(a^\dagger)^n \psi_0 = a^\dagger a a^\dagger (a^\dagger)^{n-1} \psi_0 = (a^\dagger)^n \psi_0 + (a^\dagger)^2 a (a^\dagger)^{n-1} \psi_0 = 2(a^\dagger)^n \psi_0 + (a^\dagger)^3 a (a^\dagger)^{n-2} \psi_0.$$

At this point, we can see the pattern appearing: Eventually, we will move the a all the way to the right, which will kill ψ_0 , and this will happen precisely when the first term looks like

$$n(a^\dagger)^n \psi_0.$$

f) We already know that the "lowest" solution is ψ_0 , and we know that any $\psi_n = (a^\dagger)^n \psi_0$ is a solution. There can't be any others, since if λ is not a nonnegative integer, then part (c) shows that there is no solution to the equation $N\psi = \lambda\psi$.